

Análisis de expresión diferencial - clase

Noemi Gomez

2026-09-29

Librerías

```
# install.packages("BiocManager")

# install.packages("tidyverse")
# install.packages("ggplot2")
# BiocManager::install("DESeq2")
# BiocManager::install("clusterProfiler")
# BiocManager::install("msigdb")
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.6
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2     4.0.1      v tibble     3.3.1
## v lubridate  1.9.4      v tidyr      1.3.2
## v purrr       1.2.1
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
```

```
library(DESeq2)
```

```
## Loading required package: S4Vectors
## Loading required package: stats4
## Loading required package: BiocGenerics
## Loading required package: generics
##
## Attaching package: 'generics'
##
## The following object is masked from 'package:lubridate':
##
##     as.difftime
##
## The following object is masked from 'package:dplyr':
##
```

```

##      explain
##
## The following objects are masked from 'package:base':
##
##      as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
##      setequal, union
##
##
## Attaching package: 'BiocGenerics'
##
## The following object is masked from 'package:dplyr':
##
##      combine
##
## The following objects are masked from 'package:stats':
##
##      IQR, mad, sd, var, xtabs
##
## The following objects are masked from 'package:base':
##
##      anyDuplicated, aperm, append, as.data.frame, basename, cbind,
##      colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
##      get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
##      order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
##      rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
##      unsplit, which.max, which.min
##
##
## Attaching package: 'S4Vectors'
##
## The following objects are masked from 'package:lubridate':
##
##      second, second<-
##
## The following objects are masked from 'package:dplyr':
##
##      first, rename
##
## The following object is masked from 'package:tidyr':
##
##      expand
##
## The following object is masked from 'package:utils':
##
##      findMatches
##
## The following objects are masked from 'package:base':
##
##      expand.grid, I, unname
##
## Loading required package: IRanges
##
## Attaching package: 'IRanges'
##

```

```

## The following object is masked from 'package:lubridate':
##
##   %within%
##
## The following objects are masked from 'package:dplyr':
##
##   collapse, desc, slice
##
## The following object is masked from 'package:purrr':
##
##   reduce
##
## Loading required package: GenomicRanges
## Loading required package: GenomeInfoDb
## Loading required package: SummarizedExperiment
## Loading required package: MatrixGenerics
## Loading required package: matrixStats
##
## Attaching package: 'matrixStats'
##
## The following object is masked from 'package:dplyr':
##
##   count
##
## Attaching package: 'MatrixGenerics'
##
## The following objects are masked from 'package:matrixStats':
##
##   colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
##   colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
##   colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
##   colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
##   colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
##   colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
##   colWeightedMeans, colWeightedMedians, colWeightedSds,
##   colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
##   rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
##   rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
##   rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
##   rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
##   rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
##   rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
##   rowWeightedSds, rowWeightedVars
##
## Loading required package: Biobase
## Welcome to Bioconductor
##
##   Vignettes contain introductory material; view with
##   'browseVignettes()'. To cite Bioconductor, see
##   'citation("Biobase")', and for packages 'citation("pkgname)".
##
## Attaching package: 'Biobase'

```

```
##
## The following object is masked from 'package:MatrixGenerics':
##
##     rowMedians
##
## The following objects are masked from 'package:matrixStats':
##
##     anyMissing, rowMedians
```

```
library(clusterProfiler)
```

```
##
## clusterProfiler v4.16.0 Learn more at https://yulab-smu.top/contribution-knowledge-mining/
##
## Please cite:
##
## Guangchuang Yu, Li-Gen Wang, Yanyan Han and Qing-Yu He.
## clusterProfiler: an R package for comparing biological themes among
## gene clusters. OMICS: A Journal of Integrative Biology. 2012,
## 16(5):284-287
##
## Attaching package: 'clusterProfiler'
##
## The following object is masked from 'package:IRanges':
##
##     slice
##
## The following object is masked from 'package:S4Vectors':
##
##     rename
##
## The following object is masked from 'package:purrr':
##
##     simplify
##
## The following object is masked from 'package:stats':
##
##     filter
```

```
library(msigdb)
```

```
packageVersion("tidyverse")
```

```
## [1] '2.0.0'
```

```
packageVersion("ggplot2")
```

```
## [1] '4.0.1'
```

```
packageVersion("DESeq2")
```

```
## [1] '1.48.2'
```

```
packageVersion("clusterProfiler")
```

```
## [1] '4.16.0'
```

```
packageVersion("msigdbR")
```

```
## [1] '25.1.1'
```

Cargar datos

```
rna <- read.table(file="../input/coad_rna.tsv", sep="\t", header=TRUE, check.names = FALSE, row.names =  
metadata <- read.table(file="../input/coad_metadata.tsv", sep="\t", header = TRUE)
```

```
head(metadata)
```

```
##      PATIENT_ID      SAMPLE_ID SUBTYPE AGE  SEX  OS_STATUS  
## 1 TCGA-A6-5667 TCGA-A6-5667-01    CIN  40 Female  0:LIVING  
## 2 TCGA-AY-4070 TCGA-AY-4070-01    CIN  50 Female  1:DECEASED  
## 3 TCGA-CM-6168 TCGA-CM-6168-01     GS  84 Female  0:LIVING  
## 4 TCGA-CA-6716 TCGA-CA-6716-01    CIN  65   Male  0:LIVING  
## 5 TCGA-F4-6569 TCGA-F4-6569-01    CIN  60   Male  0:LIVING  
## 6 TCGA-A6-A56B TCGA-A6-A56B-01    CIN  57   Male  1:DECEASED  
##      COUNTRY_OF_RESIDENCE  SITE_OF_BIOPSY purity ploidy wholeGenomeDuplication  
## 1                <NA>      Sigmoid colon  0.69  3.20                true  
## 2                <NA>        Cecum      0.31  4.39                true  
## 3      United States  Ascending colon  0.35  2.17                false  
## 4          Vietnam  Ascending colon  0.73  3.74                true  
## 5          Russia  Transverse colon  0.40  2.16                false  
## 6      United States  Sigmoid colon  0.72  3.21                true  
##      CMS metastasis  
## 1      CMS2      false  
## 2 no_label      false  
## 3      CMS4      false  
## 4      CMS2      false  
## 5      CMS4      false  
## 6 no_label      false
```

```
dim(rna)
```

```
## [1] 60660  300
```

Extraer el dato a comparar

```
# El modelo DESeq2 necesita una colData que tenga en las filas las muestras, y la columna con el valor  
coldata <- metadata %>% select(PATIENT_ID, wholeGenomeDuplication) %>%
```

```

column_to_rownames(var = "PATIENT_ID") %>%
mutate(wholeGenomeDuplication = as.factor(wholeGenomeDuplication))

table(coldata)

## wholeGenomeDuplication
## false  true
##    178   122

# eliminar genes sin expresion
table(rowSums(rna) > 0) # TRUE genes con expresion, FALSE genes donde en todas las muestras son 0

##
## FALSE  TRUE
## 12252 48408

rna <- rna[rowSums(rna) > 0, ]
dim(rna)

## [1] 48408   300

```

Correr análisis de expresión

```

dds <- DESeqDataSetFromMatrix(countData = rna,
                              colData = coldata,
                              design = ~wholeGenomeDuplication)

## Warning in S4Vectors:::anyMissing(runValue(x_seqnames)): 'S4Vectors:::anyMissing()' is deprecated.
## Use 'anyNA()' instead.
## See help("Deprecated")

## Warning in S4Vectors:::anyMissing(runValue(strand(x))): 'S4Vectors:::anyMissing()' is deprecated.
## Use 'anyNA()' instead.
## See help("Deprecated")

dds <- DESeq(dds)

## estimating size factors

## estimating dispersions

## gene-wise dispersion estimates

## mean-dispersion relationship

## final dispersion estimates

```

```
## fitting model and testing

## -- replacing outliers and refitting for 1599 genes
## -- DESeq argument 'minReplicatesForReplace' = 7
## -- original counts are preserved in counts(dds)
```

```
## estimating dispersions
```

```
## fitting model and testing
```

```
resultsNames(dds)
```

```
## [1] "Intercept"
## [2] "wholeGenomeDuplication_true_vs_false"
```

```
res <- results(dds, name = "wholeGenomeDuplication_true_vs_false") %>% as.data.frame()
summary(res)
```

```
##      baseMean      log2FoldChange      lfcSE      stat
## Min.      : 0.000      Min.      :-4.88285      Min.      :0.0000      Min.      :-8.76153
## 1st Qu.: 0.020      1st Qu.: -0.11598      1st Qu.: 0.1368      1st Qu.: -0.27856
## Median : 0.208      Median : -0.02934      Median : 0.4921      Median : -0.01438
## Mean    : 20.693      Mean    : -0.03760      Mean    : 1.2213      Mean     : -0.05983
## 3rd Qu.: 4.295      3rd Qu.: 0.04479      3rd Qu.: 2.9468      3rd Qu.: 0.09226
## Max.    :33423.206      Max.    : 5.25742      Max.    : 2.9468      Max.     :10.86904
##
##      pvalue      padj
## Min.      :0.0000      Min.      :0.0000
## 1st Qu.:0.2494      1st Qu.:0.1317
## Median :0.8511      Median :0.4539
## Mean     :0.6435      Mean     :0.4575
## 3rd Qu.:0.9841      3rd Qu.:0.7649
## Max.     :1.0000      Max.     :0.9999
##
##      NA's      :25371
```

```
# log2FC > 0 - enriquecido en true, < 0 enriquecido en false
# Un log2FC de 1 indica un cambio de dos veces. Es decir, si es 1 entonces en esa condicion esta expresado
true_enriched <- res %>% filter(log2FoldChange > 1, padj < 0.1)
false_enriched <- res %>% filter(log2FoldChange < -1, padj < 0.1)

cat("Hay", nrow(true_enriched), "genes sobreexpresados en duplicacion de genoma completo, versus", nrow(false_enriched), "genes subexpresados en duplicacion de genoma completo.")
```

```
## Hay 101 genes sobreexpresados en duplicacion de genoma completo, versus 246 subexpresados.
```

Volcanoplot

```

# Definir colores
res <- res %>%
  mutate(gene = rownames(res),
         significant = ifelse(gene %in% rownames(true_enriched), "WGD",
                             ifelse(gene %in% rownames(false_enriched),
                                    "noWGD",
                                    "not significant")))

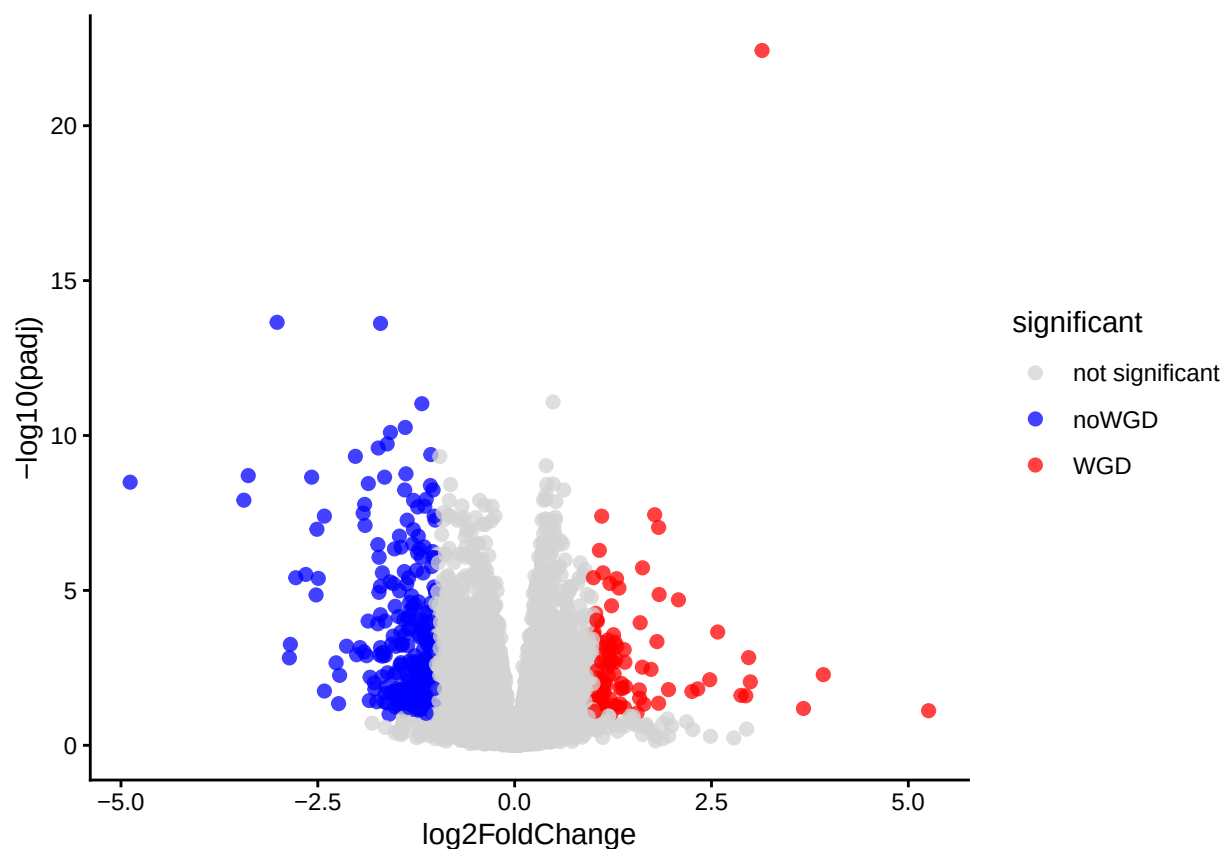
# generar volcano
ggplot(res, aes(x = log2FoldChange, y = -log10(padj),
               color = significant)) +
  geom_point(alpha = 0.75, size = 2) +
  scale_color_manual(values = c("WGD" = "red",
                                "noWGD" = "blue",
                                "not significant" = "lightgray")) +
  theme_classic()

```

```

## Warning: Removed 25371 rows containing missing values or values outside the scale range
## ('geom_point()').

```



```

# Definir una base de datos. Por ejemplo = GO procesos biológicos
go_bp <- msigdb(species = "Homo sapiens",
                 collection = "C5",
                 subcollection = "GO:BP")

```

```
# El universo de genes a estudiar debe ser los genes utilizados en la prueba
universo <- res$gene
```

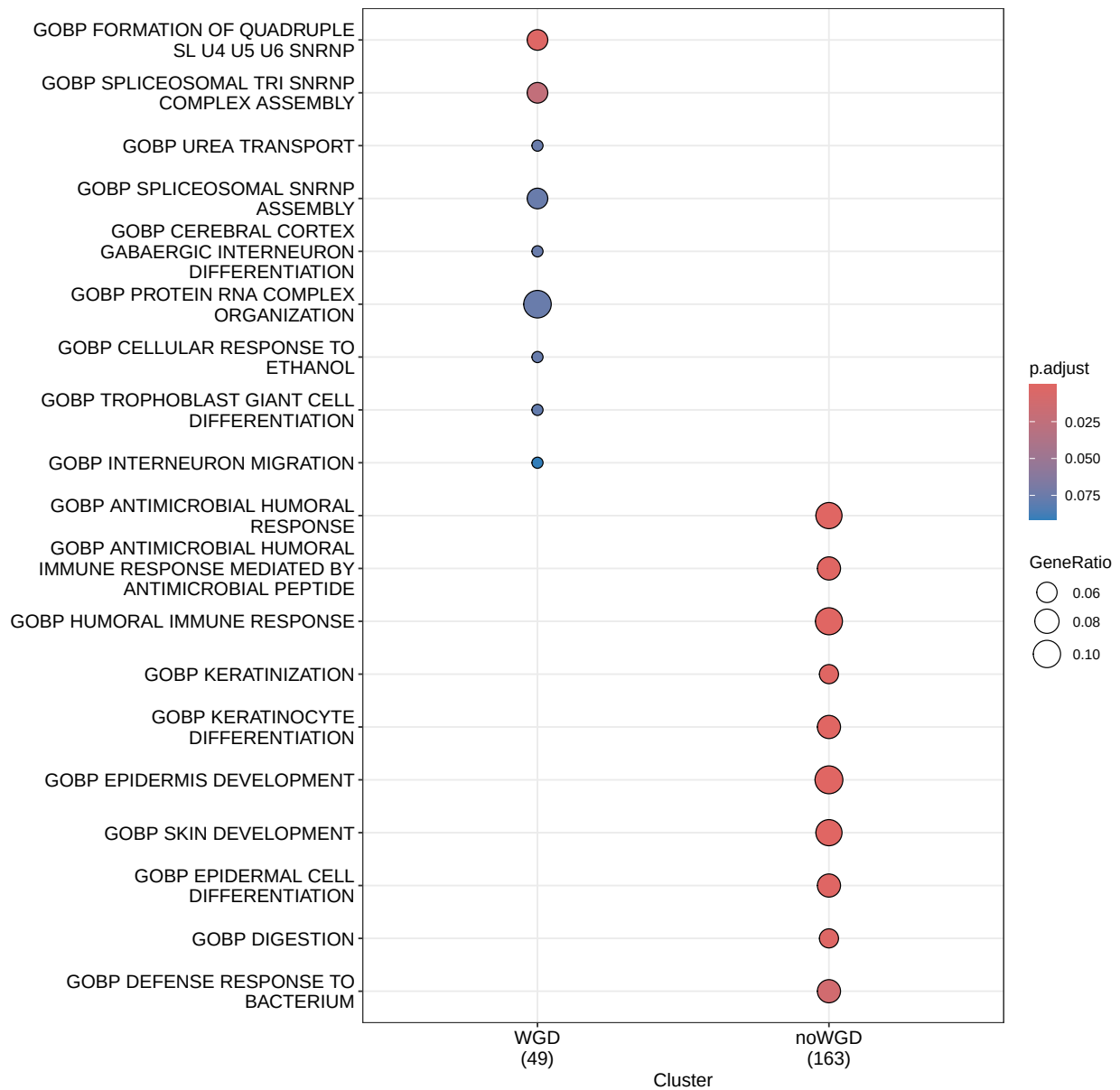
```
# Se establece una tabla de asociacion proceso-gen
term2gene <- unique(as.data.frame(go_bp[, c("gs_name", "ensembl_gene"))))
```

```
# lista de genes con grupo
genes <- list("WGD" = res$gene[res$significant == "WGD"],
             "noWGD" = res$gene[res$significant == "noWGD"])

enrich <- compareCluster(geneClusters = genes,
                        fun = "enricher",
                        TERM2GENE = term2gene,
                        universe = universo,
                        pAdjustMethod = "BH",
                        pvalueCutoff = 0.1,
                        minGSSize = 3)
```

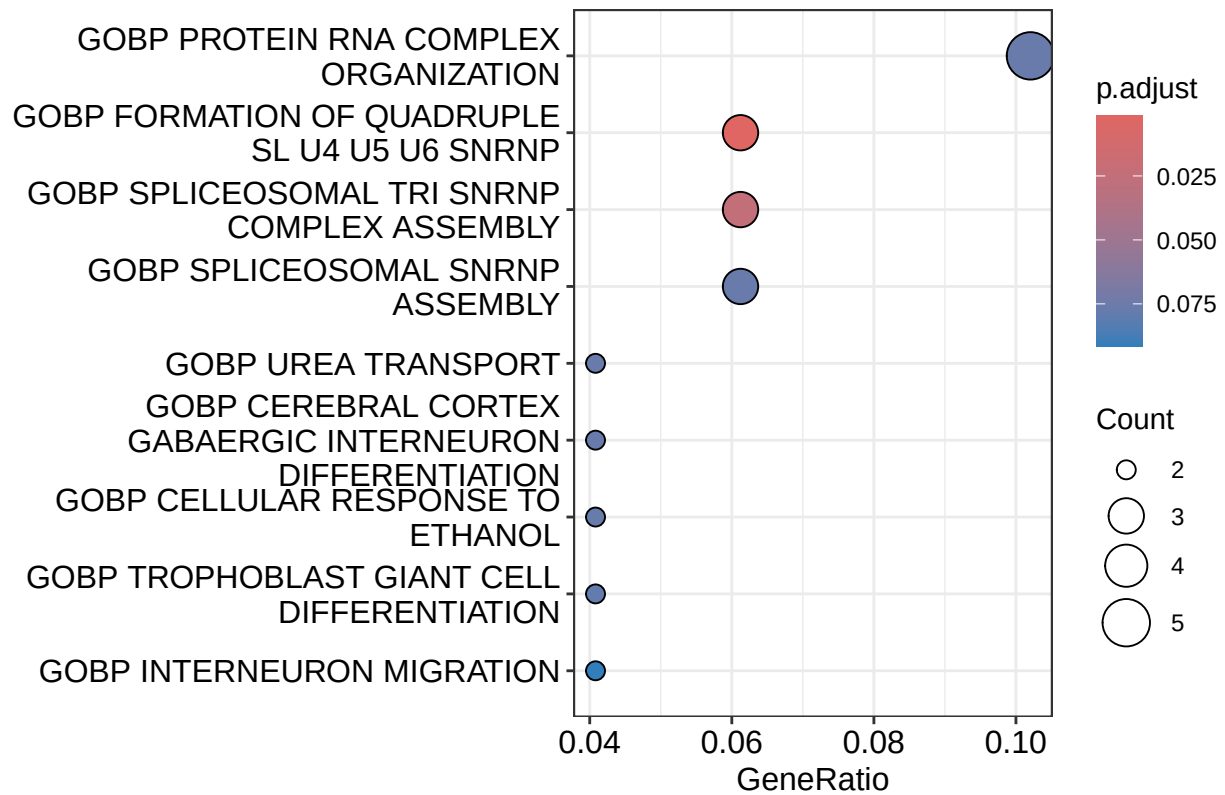
```
dotplot(enrich, showCategory = 10,
        color = "p.adjust")
```

```
## Warning: 'aes_string()' was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with 'aes()'.
## i See also 'vignette("ggplot2-in-packages")' for more information.
## i The deprecated feature was likely used in the enrichplot package.
##   Please report the issue at
##   <https://github.com/GuangchuangYu/enrichplot/issues>.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

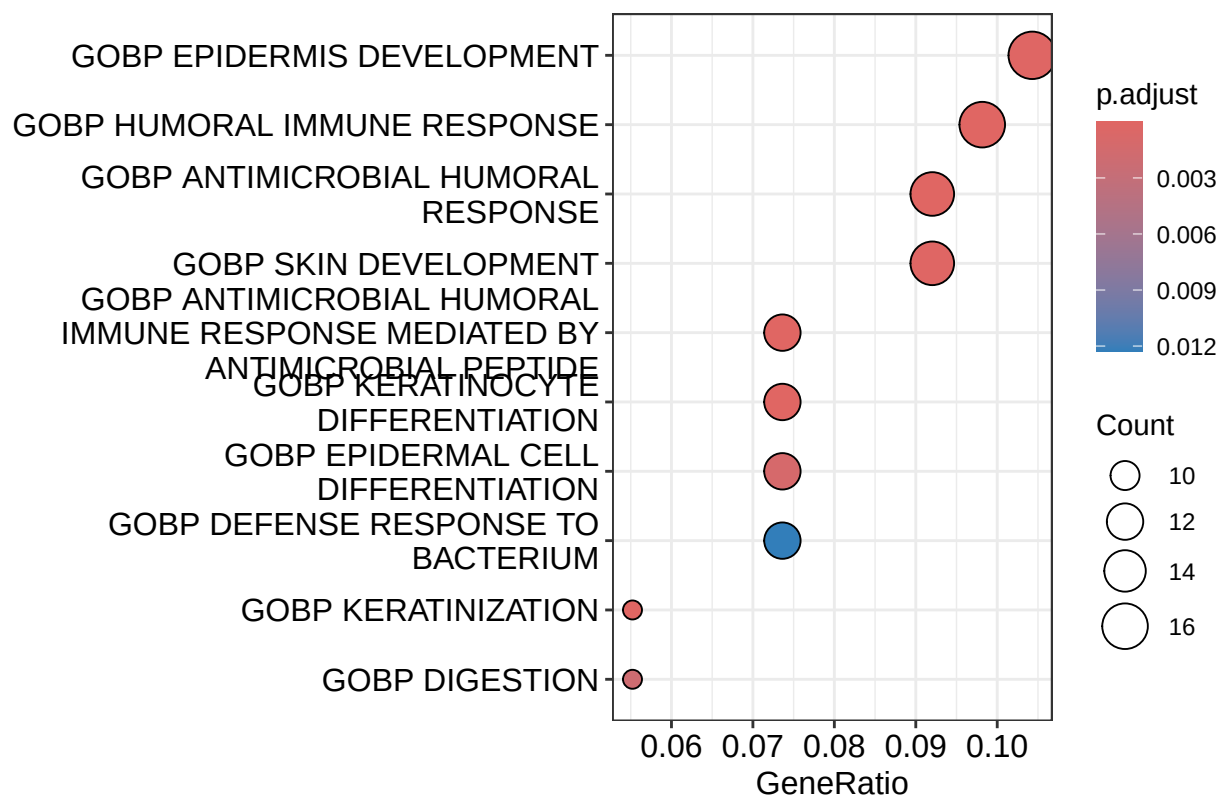


Alternativa: un enriquecimiento por categoria

```
wgd_enrich <- enricher(gene = genes$WGD, TERM2GENE = term2gene,
                        universe = universo, pAdjustMethod = "BH",
                        pvalueCutoff = 0.1, minGSSize = 3)
dotplot(wgd_enrich, showCategory = 10,
        color = "p.adjust")
```



```
nowgd_enrich <- enricher(gene = genes$noWGD, TERM2GENE = term2gene,
  universe = universo, pAdjustMethod = "BH",
  pvalueCutoff = 0.1, minGSSize = 3)
dotplot(nowgd_enrich, showCategory = 10,
  color = "p.adjust")
```



```
sessionInfo()
```

```
## R version 4.5.1 (2025-06-13)
## Platform: x86_64-pc-linux-gnu
## Running under: AlmaLinux 9.5 (Teal Serval)
##
## Matrix products: default
## BLAS: /mnt/beegfs/apps/r-4.5.1/lib64/R/lib/libRblas.so
## LAPACK: /mnt/beegfs/apps/r-4.5.1/lib64/R/lib/libRlapack.so; LAPACK version 3.12.1
##
## locale:
## [1] LC_CTYPE=en_US.UTF-8 LC_NUMERIC=C
## [3] LC_TIME=en_US.UTF-8 LC_COLLATE=en_US.UTF-8
## [5] LC_MONETARY=en_US.UTF-8 LC_MESSAGES=en_US.UTF-8
## [7] LC_PAPER=en_US.UTF-8 LC_NAME=C
## [9] LC_ADDRESS=C LC_TELEPHONE=C
## [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
##
## time zone: US/Eastern
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats4 stats graphics grDevices utils datasets methods
## [8] base
##
## other attached packages:
```

```

## [1] msigdbr_25.1.1          clusterProfiler_4.16.0
## [3] DESeq2_1.48.2            SummarizedExperiment_1.38.1
## [5] Biobase_2.68.0          MatrixGenerics_1.20.0
## [7] matrixStats_1.5.0       GenomicRanges_1.60.0
## [9] GenomeInfoDb_1.44.1     IRanges_2.42.0
## [11] S4Vectors_0.48.0       BiocGenerics_0.54.0
## [13] generics_0.1.4         lubridate_1.9.4
## [15] forcats_1.0.1          stringr_1.6.0
## [17] dplyr_1.1.4            purrr_1.2.1
## [19] readr_2.1.6            tidyr_1.3.2
## [21] tibble_3.3.1           ggplot2_4.0.1
## [23] tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] DBI_1.2.3                gson_0.1.0                rlang_1.1.7
## [4] magrittr_2.0.4          DOSE_4.2.0                otel_0.2.0
## [7] compiler_4.5.1         RSQLite_2.4.5            png_0.1-8
## [10] vctrs_0.7.0             reshape2_1.4.5           pkgconfig_2.0.3
## [13] crayon_1.5.3            fastmap_1.2.0            XVector_0.48.0
## [16] labeling_0.4.3          rmarkdown_2.30           enrichplot_1.28.4
## [19] tzdb_0.5.0             UCSC.utils_1.4.0         bit_4.6.0
## [22] xfun_0.56              cachem_1.1.0             aplot_0.2.9
## [25] jsonlite_2.0.0         blob_1.3.0              DelayedArray_0.34.1
## [28] BiocParallel_1.42.1    parallel_4.5.1          R6_2.6.1
## [31] stringi_1.8.7          RColorBrewer_1.1-3      GOSeqSim_2.34.0
## [34] assertthat_0.2.1       Rcpp_1.1.1              knitr_1.51
## [37] ggtangle_0.1.1         R.utils_2.13.0          igraph_2.2.1
## [40] splines_4.5.1          Matrix_1.7-3            timechange_0.3.0
## [43] tidyselect_1.2.1       qvalue_2.40.0           rstudioapi_0.18.0
## [46] dichromat_2.0-0.1     abind_1.4-8             yaml_2.3.12
## [49] codetools_0.2-20      curl_7.0.0             lattice_0.22-7
## [52] plyr_1.8.9            treeio_1.32.0           withr_3.0.2
## [55] KEGGREST_1.48.1       S7_0.2.1               evaluate_1.0.5
## [58] gridGraphics_0.5-1    Biostrings_2.76.0       ggtree_3.16.3
## [61] pillar_1.11.1         ggfun_0.2.0             hms_1.1.4
## [64] tidytree_0.4.7        scales_1.4.0           glue_1.8.0
## [67] lazyeval_0.2.2        tools_4.5.1            data.table_1.18.0
## [70] fgsea_1.34.2          locfit_1.5-9.12        babelgene_22.9
## [73] fs_1.6.6             fastmatch_1.1-8        cowplot_1.2.0
## [76] grid_4.5.1           ape_5.8-1              AnnotationDbi_1.72.0
## [79] nlme_3.1-168         patchwork_1.3.2        GenomeInfoDbData_1.2.14
## [82] cli_3.6.5            rappdirs_0.3.4         S4Arrays_1.8.1
## [85] gtable_0.3.6          R.methodsS3_1.8.2      yulab.utils_0.2.3
## [88] digest_0.6.39        ggrepel_0.9.6          ggplotify_0.1.3
## [91] SparseArray_1.8.1     farver_2.1.2           memoise_2.0.1
## [94] htmltools_0.5.9      R.oo_1.27.1           lifecycle_1.0.5
## [97] httr_1.4.7           GO.db_3.21.0          bit64_4.6.0-1

```